

Program6:

Implementing Clustering using K-Means Clustering Algorithm.

Aim:

To implementing Clustering using K-Means Clustering Algorithm.

Required Libraries:

pip install pandas matplotlib scikit-learn

Algorithm:

Step1: Import required libraries.

Step2: Load dataset.

Step3: Choose the number of clusters k.

Step4: Apply K-Means algorithm.

Step5: Visualize the clusters.

Step6: Evaluate with metrics like inertia and silhouette score (optional).

Source code:

```
# Step 1: Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import load_iris

# Step 2: Load a sample dataset (Iris dataset without target labels)
iris = load_iris()
X = pd.DataFrame(iris.data, columns=iris.feature_names)

# Optional: Use only first 2 features for visualization
X_vis = X.iloc[:, :2]

# Step 3: Apply KMeans clustering
k = 3 # Number of clusters
kmeans = KMeans(n_clusters=k, random_state=0)
```

```
kmeans.fit(X_vis)
```

```
# Step 4: Predict cluster labels
```

```
labels = kmeans.labels_
```

```
# Step 5: Plot the clusters
```

```
plt.scatter(X_vis.iloc[:, 0], X_vis.iloc[:, 1], c=labels, cmap='viridis', marker='o', s=100)
```

```
centers = kmeans.cluster_centers_
```

```
plt.scatter(centers[:, 0], centers[:, 1], c='red', marker='X', s=200, label='Centroids')
```

```
plt.title("K-Means Clustering")
```

```
plt.xlabel(iris.feature_names[0])
```

```
plt.ylabel(iris.feature_names[1])
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

Output:

